

A Bibliometric Mini Review of LLM-Based Multi-Agent Collaboration: Growth Dynamics, Knowledge Bases, and Future Agenda

Course Paper | A reproducible GitHub project based on 522 Lens.org records from 2020 to 2026

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Abstract

Large language models are increasingly moving beyond single-model question answering and text generation toward agentic systems that can decompose tasks, call tools, coordinate roles, and execute complex workflows. To clarify the intellectual structure and emerging trends of LLM-based multi-agent collaboration, this paper constructs a bibliometric dataset from Lens.org Scholarly Works covering 2020-2026. After DOI-based and title-based deduplication, 522 valid records were retained for analysis. We examine annual publication growth, citation impact, author collaboration, keyword co-occurrence, highly cited milestone candidates, and co-citation evidence, and then translate these bibliometric signals into a mini-review narrative.

The results indicate that the field entered a rapid expansion stage in 2024-2025, with 344 publications in 2025. The corpus received 4,265 citations in total and reached an h-index of 31. The author collaboration network suggests an active but still fragmented research community, while keyword evidence points to a four-layer thematic architecture: LLM/AI foundations, agentic collaboration architectures, RAG/RL-based capability enhancement, and governance-oriented vertical applications. Combining the Top 10 highly cited papers with thematic interpretation, the review identifies four major storylines: scientific discovery and self-driving laboratories, robotic task planning, human-AI collaboration, and medical or vertical-domain agents. The paper concludes by discussing unresolved challenges in evaluation, explainability, real-world validation, and safety governance.

Keywords: large language models; multi-agent systems; bibliometrics; keyword co-occurrence; co-citation network; mini review

1. Introduction

The development of large language models (LLMs) has reshaped the boundary of artificial intelligence systems. Early LLM applications mainly focused on question answering, summarization, translation, writing assistance, and code generation. With the rise of tool use, retrieval-augmented generation, long-context reasoning, and function calling, LLMs have increasingly been embedded into complex workflows where they can plan, retrieve, execute, evaluate, and revise.

Compared with a single LLM, a multi-agent system is not merely a larger collection of models. Its central idea is to organize roles, communication, evidence retrieval, task decomposition, and cross-checking mechanisms so that complex tasks can be handled in a more structured way. In software engineering, different agents may handle requirement analysis, architecture design, coding, testing, and review. In scientific discovery, agents may support literature search, hypothesis generation, experiment design, data analysis, and report writing. In healthcare and robotics, LLM agents may serve as high-level planners or decision-support components, but they must also be constrained by domain expertise, safety boundaries, and human oversight.

The field is expanding rapidly, which makes review writing difficult. New concepts, frameworks, and applications appear at high frequency, and the field spans artificial intelligence, robotics, software engineering, healthcare, chemistry, human-computer interaction, and safety governance. A purely narrative review may miss important structural signals. This paper therefore adopts a bibliometric mini-review approach: it uses reproducible data, figures, and representative papers to answer explicit research questions.

2. Data and Methods

2.1 Data source, query strategy, and screening

The data were retrieved from Lens.org Scholarly Works. The search strategy follows an Object AND Method logic. The Object group includes Large Language Model*, LLM*, Generative AI, Foundation Model*, and ChatGPT, while the Method group includes Multi-agent*, Collaboration, Coordination, Cooperation, and Multi-agent system*. The time window was set to 2020-2026. Document types were limited to articles, reviews, and conference papers, and the language was limited to English. The query configuration is stored in config/query.yaml. The query log and the raw export file both indicate March 26, 2026 as the project record date, which is used here as the retrieval/export date.

The initial Lens.org export contained 542 records. The dataset was deduplicated first by DOI and then by title, resulting in 522 valid records. In the cleaned dataset, Title and Publication Year have 100% coverage, Abstract has 98.28% coverage, References has 81.99% coverage, and DOI has 99.81% coverage. These fields support annual trend analysis, keyword co-occurrence, author collaboration, citation analysis, co-citation mapping, and milestone paper identification.

Transparency item	Implementation in this paper
Database	Lens.org Scholarly Works
Full query	Object AND Method; see config/query.yaml
Retrieval date	2026-03-26, based on query log and raw export record
Time window	2020-2026; 2026 is treated as partial-year evidence
Document types	Article, Review, Conference Paper
Inclusion criteria	Records jointly related to LLM/generative AI and multi-agent/collaboration topics
Exclusion criteria	Duplicates, clearly irrelevant records, or unusable core metadata
Deduplication	DOI first, then title; 542 records reduced to 522
Tools	Python project using pandas, networkx, matplotlib, Pillow; dependencies in requirements.txt
Key parameters	Keyword co-occurrence threshold ≥ 2 ; author-collaboration display threshold ≥ 2 ; Top 10 highly cited papers

2.2 Research questions and analytical framework

Following the course requirement, figures are not treated as isolated visual outputs. Each figure or table is linked to a research question. RQ1 asks how the field has grown and which stage it has entered. RQ2 asks who participates in the field and whether stable collaboration groups have formed. RQ3 asks what the major research themes are and how they are structurally related. RQ4 asks which highly influential papers constitute the knowledge base. RQ5 synthesizes the evidence to discuss future challenges.

The analysis includes annual publication trends, citation impact indicators, keyword co-occurrence, author collaboration, highly cited paper identification, co-citation analysis, and parameter sensitivity checks. The interpretation follows a Claim-Evidence-Reasoning logic: each main claim is supported by figures, indicators, or representative papers, and the reasoning explains both what can and cannot be inferred from the evidence.

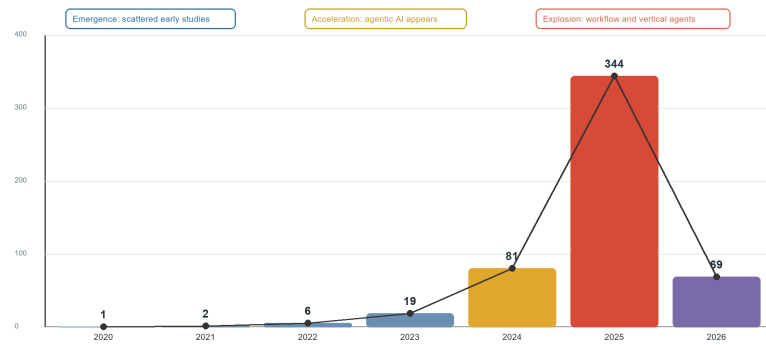
3. Bibliometric Results

3.1 Annual growth: rapid expansion after 2024

Annual publication counts reveal a clear stage transition. The dataset contains 1 paper in 2020, 2 in 2021, 6 in 2022, 19 in 2023, 81 in 2024, and 344 in 2025. The 2026 count is 69, but it should not be compared directly with complete calendar years. This pattern supports the claim that LLM-based multi-agent collaboration has moved from early conceptual exploration to rapid expansion, but it does not by itself prove that the technology is already mature.

Fig. 2. Publication Growth and Development Stages

Annual output of LLM multi-agent collaboration research, 2020-2026



Caption: Publication output rises sharply after 2024, indicating a transition from early exploration to high-intensity expansion. The 2026 bar is incomplete and should not be compared with full years.

Figure 1. Annual publication growth and development stages. Data source: Lens.org Scholarly Works; time window: 2020-2026; unit of analysis: scholarly work; tool: Python. Bar height indicates annual publication count and color marks exploratory, accelerating, and explosive stages. The 2026 data are partial-year observations..

3.2 Author collaboration: active participation but fragmented stable ties

The author collaboration network addresses who is working on this topic and whether stable collaboration groups have emerged. The corpus includes 2,564 authors, with an average of 5.07 authors per paper. After filtering one-off coauthorships, the displayed network contains 51 nodes and 48 edges. This suggests that the field is active and expanding, but repeated collaboration is not yet highly concentrated around a small number of core teams.

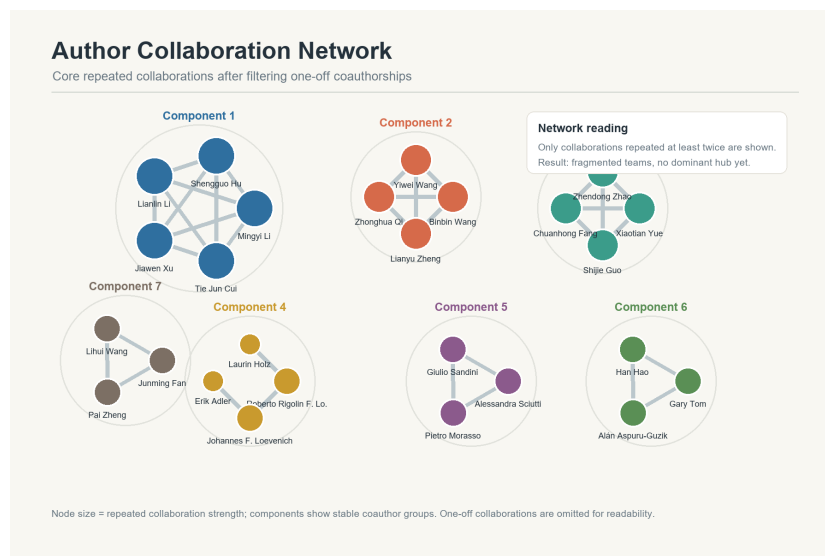


Figure 2. Author collaboration network. Data source: 522 cleaned Lens.org records from 2020-2026; unit of analysis: author; edges represent coauthorship within the same paper; display threshold: at least two repeated collaborations. Node size reflects repeated collaboration strength. The figure answers RQ2 about collaboration structure..

3.3 Keyword structure: a four-layer thematic architecture

The most frequent keywords include artificial intelligence, large language models, machine learning, natural language processing, generative AI, agentic AI, multi-agent systems, retrieval-augmented generation, AI agents, human-robot interaction, and digital health. These keywords form a layered structure rather than a random list of topics.

Figure 3 translates the keyword evidence into a review framework. The first layer is the LLM/AI foundation, which provides language understanding, generation, reasoning, and knowledge representation. The second layer is the agentic collaboration architecture, involving role allocation, task decomposition,

communication protocols, and reflection. The third layer includes RAG and reinforcement learning as capability-enhancement mechanisms. The fourth layer covers governance and vertical applications such as ethics, human-robot interaction, digital health, robotics, and medicine.

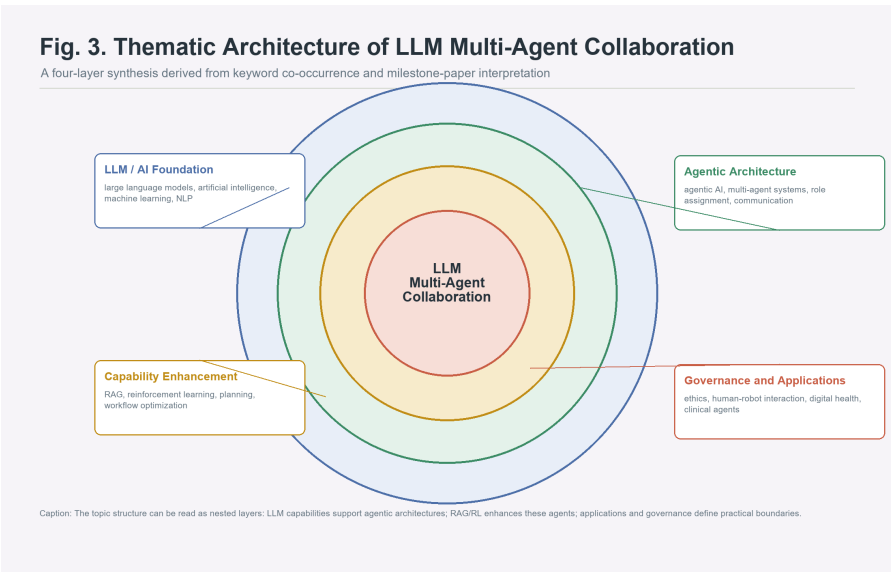


Figure 3. Thematic architecture of LLM-based multi-agent collaboration. Data source: cleaned Lens.org corpus, 2020-2026; unit of analysis: keyword; within-paper keywords are deduplicated; co-occurrence threshold: at least two occurrences. The layers summarize how technical foundations, collaboration architectures, capability enhancement, and applications are connected.

3.4 Highly cited papers: from a paper list to a knowledge base

The required table in a bibliometric review should support the argument rather than merely list papers. We rank papers by Lens.org Citing Works Count and select the Top 10 as first-pass milestone candidates. Table 1 indicates that influential works are concentrated around scientific discovery and self-driving laboratories, robotic task planning, human-AI collaboration, and medical or vertical-domain agents.

Table 1. Top 10 highly cited milestone candidates. Data source: cleaned Lens.org records from 2020-2026; unit of analysis: paper; ranking metric: Citing Works Count; threshold: Top 10.

Rank	Year	Cites	Theme	Representative paper
1	2024	352	Scientific discovery	Self-Driving Laboratories for Chemistry and Materials Science.
2	2023	259	Robotics / planning	A Comprehensive Review of Recent Research Trends on Unmanned Aerial V...
3	2023	173	Human-AI collaboration	Co-Writing Screenplays and Theatre Scripts with Language Models: Eval...
4	2025	136	Scientific discovery	A review of large language models and autonomous agents in chemistry.
5	2024	109	Robotics / planning	SMART-LLM: Smart Multi-Agent Robot Task Planning using Large Language...
6	2024	108	Medical agents	Evaluating large language models as agents in the clinic.
7	2023	104	Human-AI collaboration	Defining human-AI teaming the human-centered way: a scoping review an...
8	2024	101	Robotics / planning	RoCo: Dialectic Multi-Robot Collaboration with Large Language Models
9	2024	100	Robotics / planning	Understanding Large-Language Model (LLM)-powered Human-Robot Interaction
10	2024	90	Medical agents	The impact of large language models on radiology: a guide for radiolo...

Citation counts should be interpreted cautiously. A highly cited paper is not automatically correct, nor does it prove that a technical path has matured. It is better treated as an anchor for identifying knowledge bases and representative storylines. Therefore, this review uses Table 1 together with keyword and co-citation evidence rather than as a simple ranking.

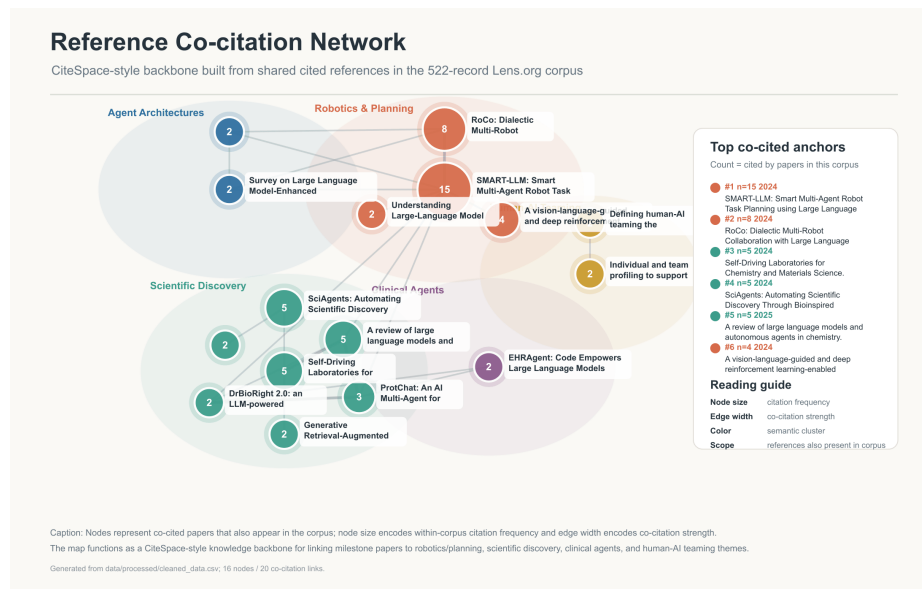


Figure 4. Reference co-citation network. Data source: cleaned Lens.org corpus from 2020-2026; unit of analysis: co-cited representative references. Node size indicates citation frequency within the corpus, edges indicate co-citation strength, and colors indicate thematic clusters. The figure complements Table 1 by showing shared knowledge bases behind highly cited papers..

4. Discussion: storylines, limitations, and future agenda

The evidence supports four major storylines. The first is multi-agent task planning and automated workflows, where LLM agents transform natural-language goals into executable subtasks. The second is scientific discovery and self-driving laboratories, where agents can connect literature search, hypothesis generation, experiment design, and result analysis. The third is human-AI collaboration and human oversight, which reframes multi-agent systems as collaborations among humans, models, tools, and environments. The fourth is medical, robotic, and vertical-domain applications, where reliability, auditability, and domain constraints become essential.

Several challenges remain. First, evaluation is not yet standardized: many systems demonstrate task completion but lack comparable benchmarks, failure-recovery metrics, and cost measures. Second, collaboration mechanisms remain difficult to explain; when a multi-agent system fails, it is often unclear whether the error comes from retrieval, reasoning, communication, or tool execution. Third, vertical scenarios require stronger real-world validation by domain experts. Fourth, safety and ethical issues are more complex in multi-agent settings because multiple agents may amplify each other's errors or blur responsibility boundaries.

This review has limitations. It uses Lens.org as the sole database, so the results may be influenced by database coverage and metadata quality. The 2026 data are incomplete and should not be compared directly with full years. References coverage is 81.99%, which supports an initial co-citation analysis but may miss some citation relations. Finally, the milestone candidates are based primarily on Lens.org citation counts; future work can integrate burst detection, betweenness centrality, cluster position, and manual full-text reading.

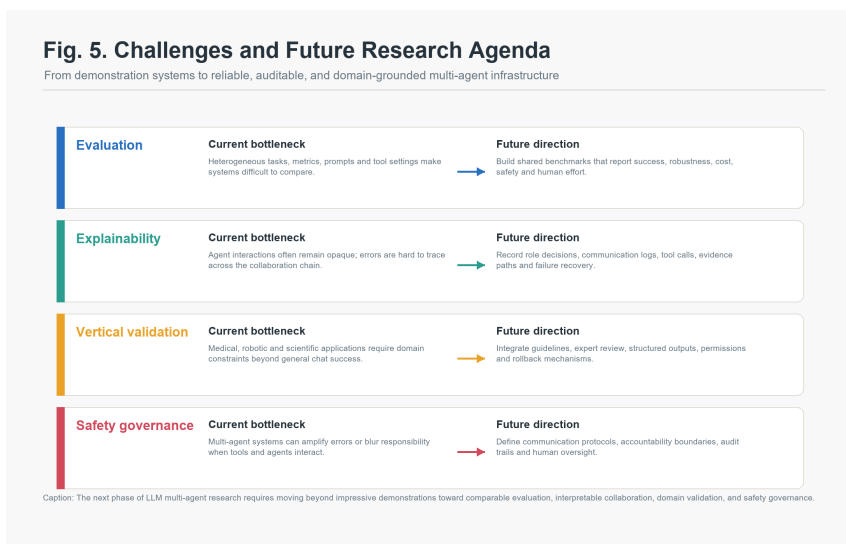


Figure 5. Challenges and future research agenda. This figure synthesizes RQ1-RQ4 into four future directions: evaluation, collaboration mechanisms, vertical-scenario validation, and safety governance. Arrows indicate how current bottlenecks are translated into research agenda items.

5. Conclusion

Based on 522 Lens.org records, this paper provides a bibliometric mini review of LLM-based multi-agent collaboration. The field entered rapid expansion in 2024-2025, and its thematic structure has moved beyond single-model capability toward agentic collaboration architectures, RAG/RL-based enhancement, governance issues, and vertical applications. The author collaboration network suggests an expanding but still fragmented community, while highly cited papers and co-citation evidence help identify scientific discovery, robotic planning, human-AI collaboration, and medical agents as major knowledge bases.

The value of LLM-based multi-agent collaboration is not simply the addition of more agents. Its value lies in embedding LLMs into more controllable workflows through role division, evidence retrieval, cross-checking, and human oversight. Future research should move from demonstration systems toward systems that are evaluable, explainable, auditable, and deployable under real constraints.

AI Use Statement and Project Materials

AI assistance was used for structure checking, language polishing, figure-interpretation logic, and code troubleshooting. It was not used to fabricate references, replace human judgment, or generate unverifiable data conclusions. All core data, figures, tables, and claims are traceable to the GitHub project and were manually checked. The detailed statement is stored in docs/ai_usage.md.

Key project materials include config/query.yaml, data/processed/cleaned_data.csv, outputs/metrics_summary.txt, outputs/keyword_statistics.txt, outputs/tables/top_cited_papers.md, outputs/figures/author_collaboration_network.png, outputs/review_figures/fig2_publication_growth_stages.png, outputs/review_figures/fig3_thematic_architecture.png, params.md, and README.md.

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